UM's Bicycle Rental Management System

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CISC3003 Web Programming | Team 01

Contribution Scope

Data, Operations, & Deployment

Focusing on the system's foundation layer to ensure reliability, manageability, and reproducible initialization workflows.

WORK BREAKDOWN

Core Foundation

Database architecture designed for absolute state consistency across the rental lifecycle.

Operations

Admin and Staff modules featuring a secure write-back loop and permission layering.

Deployment

Automated reproducible initialization for standardized team testing and delivery.

THE FOUNDATION: STATE CONSISTENCY

Rental systems risk **State Inconsistency**: orphaned orders, ghost vehicles, or settlement errors.

- Unified model for users, vehicles, and stations.
- Integrated wallet transactions and audit logs.
- Verifiable state transitions at every operation.

DATABASE SCRIPT LAYER

schema.sql

Defines core entities: vehicles, orders, and wallets. Enforces relational integrity via foreign keys.

seed_demo.sql

Initializes demo-ready data: test accounts, available vehicles, and baseline operational records.

reset_state.sql

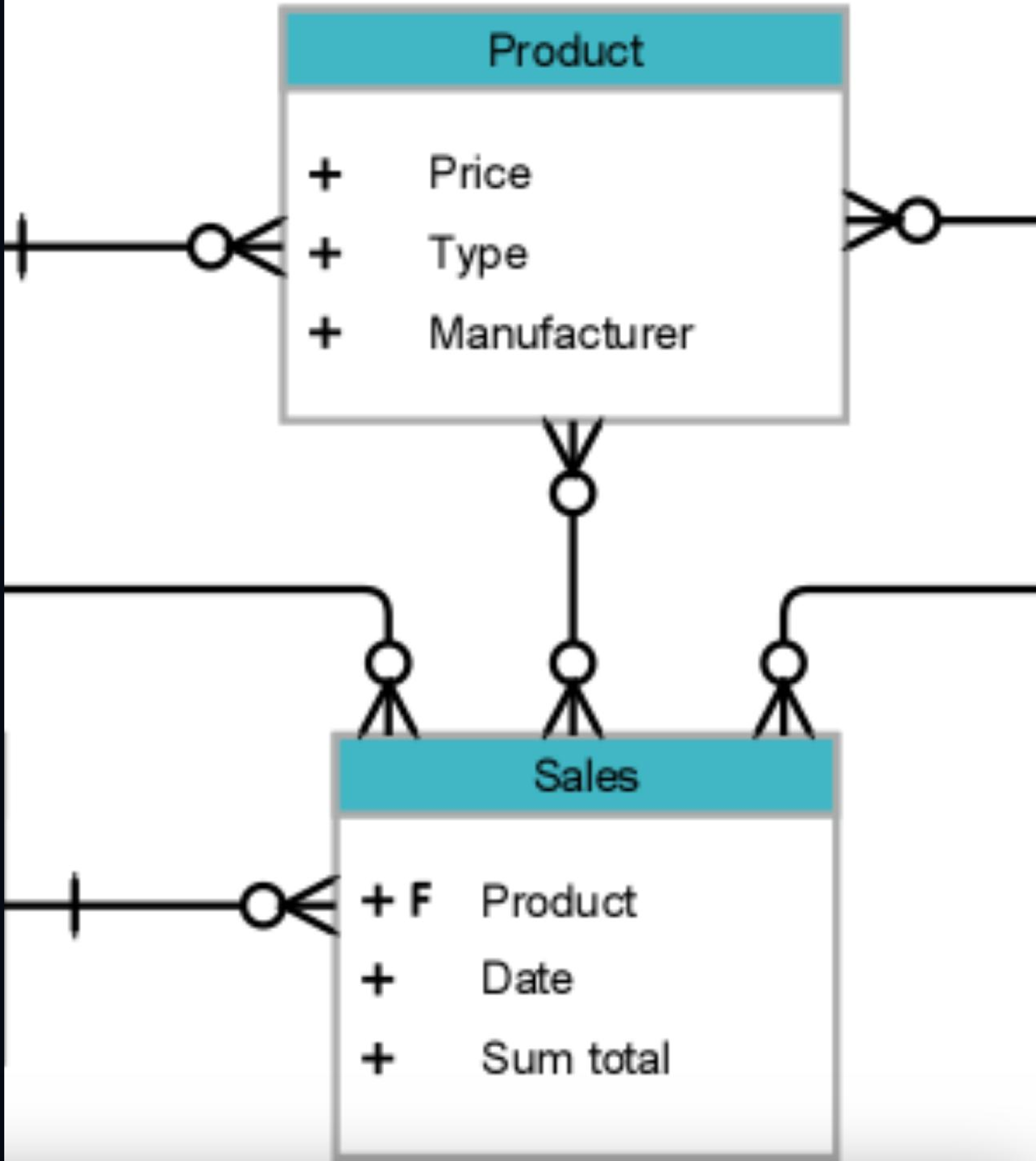
Clears environment residues before demonstrations to ensure repeatable test conditions.

SYSTEM BLUEPRINT

Relational Integrity

The system is modeled as a verifiable data model where every entity transition—from rental start to fee settlement—is tracked.

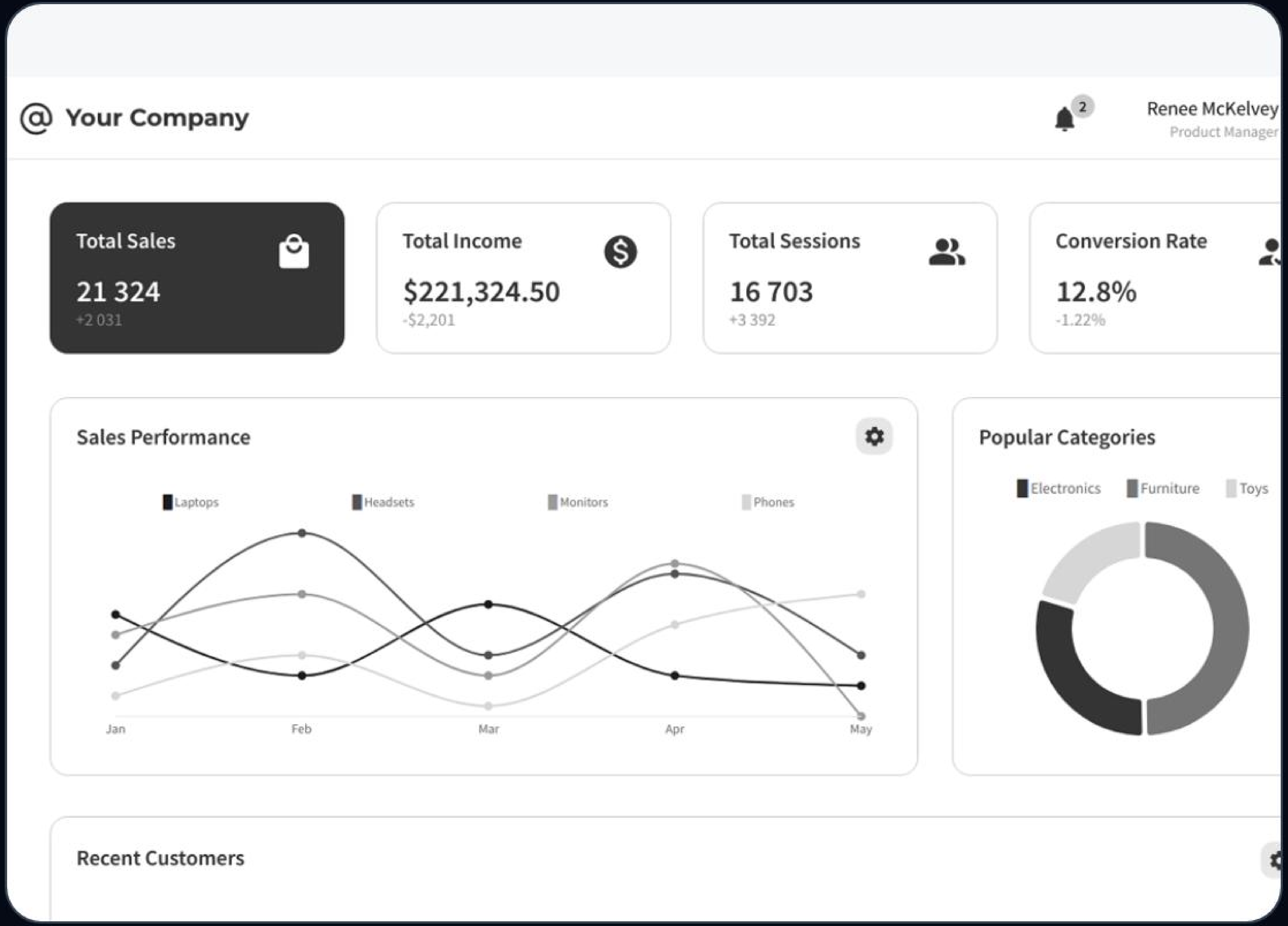
Indexes on high-frequency columns support a scalable dashboard and efficient real-time monitoring.



Management & Operations

Admin/Staff Modules and the Operational Write-Back Loop

ADMINISTRATIVE CONTROL CENTER



Real Operational Loop

Management is not just UI; it's a backend-driven loop:

admin_dashboard.php: Oversight of all system states.

staff_dashboard.php: Station-level operational controls.

rental_action.php: Core action dispatcher for validated DB updates.

BACKEND SECURITY LAYER

Permission Layering

Staff and admin actions are role-guarded before any database mutation occurs.

Write Consistency

Direct synchronization between UI state and backend DB records for high reliability.

Traceability

Every management action is recorded in audit logs for operational review.

DEPLOYMENT EFFICIENCY

100%

Repeatable Setup

Reliable Delivery Chain

By standardizing the environment, we reduce "it works on my machine" errors and ensure instructors see exactly what we intended.

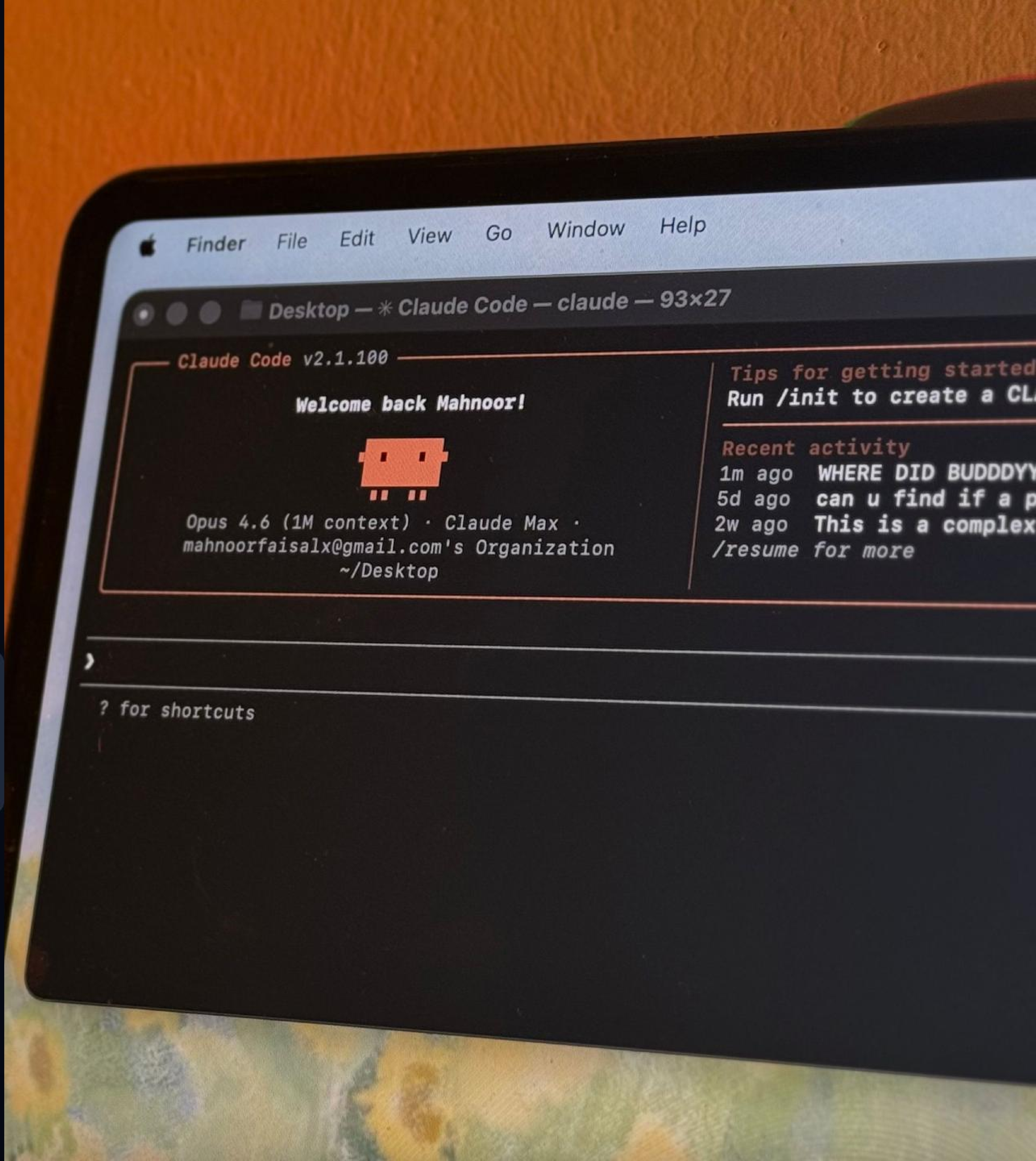
- Automated SQL imports
- Batch script execution
- Standardized data baseline

DEMO RELIABILITY

setup_database.bat

One-command initialization that imports schema, migrations, and seed data based on environment modes.

```
> CMD: setup_database.bat
[INFO] Importing Schema... OK
[INFO] Seeding Demo Data... OK
```



Questions?

Thank you for your attention.

Hoi Kai Cheng (Thomas)

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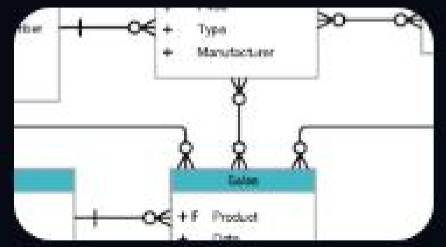
IMAGE SOURCES

https://www.augusta.edu/campus-recreation/images/rental2.jpg

Source: www.augusta.edu



https://miro.medium.com/1*ko_eoKa0I-M36IFGKERa9A.png

Source: medium.com

https://landing.moqups.com/img/templates/wireframes/admin-dashboard-wireframe.png

Source: moqups.com

https://static0.xdaimages.com/wordpress/wp-content/uploads/wm/2026/04/claude-code-terminal-window-on-mac-laptop-displaying-welcome-message-and-recent-activity.jpeg

Source: www.xda-developers.com